

KOKOU MAWULOM ADJE

AI ENGINEER | MACHINE LEARNING ENGINEER | COMPUTER VISION ENGINEER

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PROFESSIONAL SUMMARY

AI and machine learning engineer with an M.S. in Artificial Intelligence, working across computer vision, deep learning, and LLM evaluation. Fine-tuned a YOLOv8 detector to 0.983 mAP@0.5 and exported it to ONNX, and benchmarked classical and deep models on real-world data. Contract experience evaluating LLMs through a remote AI fellowship.

TECHNICAL SKILLS

Languages: Python, Java, C#, SQL

ML & Deep Learning: PyTorch, TensorFlow, scikit-learn, Hugging Face, YOLOv8, ONNX Runtime, OpenCV, Albumentations

Areas: Computer Vision, NLP, LLM training & evaluation, transfer learning, model benchmarking

Data & Tooling: NumPy, Pandas, Matplotlib, Streamlit, Git, Linux, PostgreSQL, MySQL, MongoDB

EXPERIENCE

AI Trainer – Machine Learning & Computer Vision Expert

Oct 2025 – May 2026

Handshake AI Fellowship | Remote

- Developed and evaluated domain-specific prompts to assess the performance of large language models (LLMs) and AI systems in machine learning and computer vision.
- Analyzed AI model outputs for accuracy, clarity, and depth across specialized subfields including computer vision, NLP, and audio/speech.
- Contributed to improving AI system performance through iterative prompt development, output evaluation, and structured feedback across machine learning and computer vision tasks.
- Conducted independent research to support prompt development and model evaluation tasks.

PROJECTS

Vision-Scan — PCB Defect Detection (YOLOv8)

github.com/Kokou-Adje/vision-scan

- Fine-tuned YOLOv8s from COCO weights to 0.983 mAP@0.5 at 99 FPS, detecting six manufacturing-defect classes on the DeepPCB benchmark (single NVIDIA T4 GPU).
- Built a three-stage pipeline: bounding-box-aware augmentation (~4x on 1,050 images), FP16 mixed-precision training with AdamW and cosine LR decay, and ONNX export for CPU/CUDA/OpenVINO inference.
- Tech:** PyTorch, Ultralytics YOLOv8, Albumentations, ONNX Runtime, OpenCV.

Network Intrusion Detection — Model Benchmark

github.com/Kokou-Adje/cyber-intrusion-detection

- Benchmarked a Random Forest against a 1D-CNN and Transformer for network-attack detection on CIC-IDS2017 (76 features, 9 classes, 378K test flows); the Random Forest won at 99.8% accuracy with a 0.001 attack false-positive rate.
- Evaluated with per-class recall and macro-F1 instead of raw accuracy, weighted classes to learn rare attacks (as few as 11 samples), and reported the Transformer's failure as an honest negative result.
- Tech:** Python, TensorFlow, scikit-learn, Streamlit.

Pneumonia Detection from Chest X-Rays (DenseNet-121)

github.com/Kokou-Adje/pneumonia-detection-densenet121

- Built a DenseNet-121 transfer-learning model that classifies chest X-rays as normal or pneumonia, reaching ~88% test accuracy and 0.95 ROC-AUC with balanced sensitivity and specificity.
- Applied class balancing for the uneven class distribution and deployed the model as a Streamlit web app that returns a live prediction from an uploaded X-ray.
- Tech:** TensorFlow, Keras, DenseNet-121, Streamlit, NumPy.

EDUCATION

M.S. Artificial Intelligence — Kennesaw State University

May 2026

Coursework: Machine Learning, Deep Learning, Machine Vision, NLP, Reinforcement Learning, Data Mining.

B.S. Computer Science — Kennesaw State University

May 2024

Dean's List (Spring 2024), President's List (Summer 2023).

A.A.S. Computer Science — Chattahoochee Technical College

May 2021

CERTIFICATIONS

- IBM Machine Learning Professional Certificate — IBM (In Progress)
- Java Programmer Certificate — Chattahoochee Technical College
- C# Programmer Certificate — Chattahoochee Technical College